

Early Sepsis Prediction

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Introduction

Sepsis is a life-threatening condition that occurs when the body's response to infection causes tissue damage, organ failure, or death. In the U.S. alone, 1.7 million cases and 350,000 deaths occur each year ([CDC](#)), accounting for \$24 billion annually ([Paoli et al., 2018](#)). Early detection is critical, as each hour of delayed treatment increases mortality by 4–9% ([Seymour et al., 2017](#)). The goal of this project is to develop a deep learning predictive model to identify sepsis in ICU patients before clinical diagnosis, enabling earlier intervention to reduce mortality, decrease treatment costs, and improve resource allocation.

Stakeholders and KPIs

Our primary stakeholders—patients, hospital systems, and clinical teams—require a solution that balances clinical reliability with operational costs. Given that missed sepsis diagnoses carry fatal consequences while false positives affect resource utilization, we prioritized sensitivity alongside precision, with AUROC as a general performance metric. We require at least **15% improvement in sensitivity** over non-deep learning baselines along with quick inference time to be considered feasible for implementation.

Dataset and Preparation

The dataset consists of 40,000+ ICU patients' hourly time-series data across 40 clinical variables (vital signs, lab results) and demographics. Each patient was given a label identifying whether or not the patient was later diagnosed with sepsis. After handling missing values through interpolation, standardizing features, and removing sparse variables, we maintained sepsis prevalence across training, validation, and test set stratification.

Models and Results

We established a logistic regression baseline to benchmark deep learning performance. While quick to train on the large dataset, the model did not achieve high accuracy metrics, with a test AUROC of 66% and low precision/recall scores. At this time, it is clear that the baseline model is insufficient for clinical deployment and further work is needed to address issues impacting performance.

Three deep learning architectures were engineered and their performances compared: GRU-D, TCN, and TRT. A Gated Recurrent Unit with Decay (GRU-D) is a deep-learning model tailored for irregular time-series data with lots of missing values. It learns both temporal trends and missingness patterns, enabling detection of rare events due to class imbalance. Temporal Convolutional Networks (TCNs), by contrast, use one-dimensional causal convolutions to process sequences in parallel while maintaining the correct temporal order, offering faster training and better stability for longer input sequences. Finally, a Temporal Residual Transformer (TRT) model applies the transformer architecture's attention mechanism to time-series data, allowing the model to dynamically weigh the importance of past observations rather than relying on fixed memory or local convolutional windows. Together, these models span the major paradigms for time-based learning—recurrent, convolutional, and attention-based—providing a wide span of options.

Ultimately, we selected the **Temporal Residual Transformer (TRT)** as the final model. Achieving an overall **test accuracy of 83.5%** and an **AUROC of 88%**, this model demonstrates strong generalization and discriminative ability. While its recall was high at 76%, the precision of 27% indicates some room for improvement in the confidence of its sepsis diagnosis in order to preserve hospital resources.

Future Work

Future work would aim to predict not just sepsis diagnosis at the patient-level, but the exact hour it occurs. Comparing hospitals A and B may reveal differences linked to their clinical practices. Finally, refining model weights and improving overall accuracy will further strengthen model performance.